

DESIGN AND SIMULATION OF AN LM555 ASTABLE MULTIVIBRATOR

College Project Documentation

Using KiCad EDA and Ngspice

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1. Abstract

This project presents the design, simulation and PCB implementation of an LM555 astable multivibrator. The 555 timer is configured as a free-running oscillator. R1, R2 and C1 form the timing network, while the output drives an LED through a 330 Ω current-limiting resistor. The circuit was developed in KiCad, simulated with ngspice using an external NE555 SPICE model, transferred to PCB layout, routed, verified with DRC, viewed in 3D, and prepared with Gerber and drill files.

2. Objectives

- Design an LM555 astable multivibrator using a 5 V supply.
- Simulate the circuit using KiCad/ngspice.
- Observe the time-domain output.
- Design a practical through-hole PCB.
- Assign footprints, route tracks and verify the board with DRC.
- Generate Gerber and drill files for possible fabrication.

3. Software and Tools Used

- KiCad EDA — schematic, simulation, PCB layout, DRC, 3D view and Gerber generation.
- Ngspice — SPICE simulation engine.
- External NE555 SPICE model — timer model used by the simulation.

4. Components

Reference	Component	Value / Type
U101	LM555xN / NE555 timer	DIP-8
R101	Resistor	1 k Ω
R102	Resistor	47 k Ω
R103	LED resistor	330 Ω
C101	Timing capacitor	10 μ F
C102	Control bypass capacitor	10 nF
D101	LED	5 mm THT
J1	Power connector	2-pin, +5 V/GND

5. Circuit Configuration

The LM555 is used in astable mode. Pins 2 (TRIG) and 6 (THR) share the timing node. R1 connects +5 V to the discharge node, R2 connects the discharge node to the timing node, and C1 connects the timing node to ground. Pin 4 (RESET) and pin 8 (VCC) are connected to +5 V; pin 1 is grounded. Pin 5 (CV) is bypassed with C2 = 10 nF. Pin 3 (OUT) drives the LED through R3 = 330 Ω .

6. Pin Connections

Pin	Function	Connection
1	GND	Ground
2	TRIG	Timing node with Pin 6
3	OUT/Q	330 Ω $\rightarrow$ LED $\rightarrow$ GND
4	RESET	+5 V
5	CV	10 nF $\rightarrow$ GND

6	THR	Timing node with Pin 2
7	DIS	R1/R2 junction
8	VCC	+5 V

7. Working Principle

1. C1 charges through R1 and R2.
2. When the capacitor reaches the upper threshold, the 555 changes state and the discharge transistor turns on.
3. C1 discharges through R2.
4. When the capacitor reaches the lower threshold, the 555 changes state again.
5. The charge/discharge process repeats and produces a periodic output at Pin 3.
6. The LED indicates the output switching.

8. Design Calculation

For an ideal 555 astable multivibrator:

$$T = 0.693 (R1 + 2R2) C1$$

$$f = 1/T$$

With $R1 = 1 \text{ k}\Omega$, $R2 = 47 \text{ k}\Omega$ and $C1 = 10 \text{ }\mu\text{F}$:

$$T \approx 0.659 \text{ s}$$

$$f \approx 1.52 \text{ Hz}$$

Thus the theoretical oscillation frequency is approximately 1.5 Hz. The exact simulated result can vary with the selected SPICE model and startup conditions.

9. SPICE Simulation

The circuit was simulated in KiCad using ngspice. An external NE555 model was loaded and its eight pins were mapped to the LM555xN symbol. The LED was represented by a diode SPICE model and a 5 V DC source was used. Transient analysis was used to observe the time-domain behavior.

- Transient time step: 1 ms
- Final simulation time: 2 s
- Startup workaround used during testing: UIC / initial-condition based transient startup

10. PCB Design Procedure

7. Assign footprints to the components.
8. Update the PCB from the schematic.
9. Place the components and keep timing components close to the 555.
10. Create a closed rectangular Edge.Cuts outline.
11. Route all PCB connections.
12. Add a physical 2-pin +5 V/GND connector.
13. Run DRC and correct unconnected items.
14. Inspect the completed board in 3D.

11. PCB Verification

The final PCB was checked with KiCad Design Rules Checker. The completed design reached zero violations and zero unconnected items. The PCB was also inspected in the 3D Viewer to verify the board outline and component placement.

12. Manufacturing Files

Gerber and Excellon drill files were generated from the completed PCB. The manufacturing package contains the copper, solder-mask, silkscreen, Edge.Cuts and drill information.

- F.Cu — front copper
- B.Cu — back copper
- F.Mask — front solder mask
- B.Mask — back solder mask
- F.Silkscreen — front markings
- B.Silkscreen — back markings
- Edge.Cuts — board outline
- Excellon drill file — through-hole drill locations

13. Applications

- LED flasher and indicator circuits.
- Clock and pulse generation.
- Timing and delay circuits.
- Simple oscillator applications.
- Educational timing and digital-circuit experiments.

14. Advantages

- Simple and low-cost implementation.
- Few external components.
- Easy to simulate and prototype.
- Useful for learning oscillator and PCB-design fundamentals.
- Timing can be changed by changing R1, R2 or C1.

15. Limitations

- Frequency depends on component tolerances.
- The LED is an indicator and is not intended for high-power loads directly.
- Exact simulation behavior depends on the selected NE555 SPICE model.

16. Conclusion

The LM555 astable multivibrator was successfully developed through a complete KiCad workflow: schematic design, SPICE simulation, footprint assignment, PCB layout, routing, DRC verification, 3D inspection and manufacturing-file generation. The final PCB design achieved zero DRC violations and zero unconnected items and was prepared for possible fabrication.

17. Suggested Figures

- Figure 1 — LM555 astable schematic.
- Figure 2 — SPICE transient waveform.
- Figure 3 — Footprint assignment.
- Figure 4 — Routed PCB layout.
- Figure 5 — DRC result showing zero violations.
- Figure 6 — Final PCB 3D view.
- Figure 7 — Gerber/drill generation.

18. Final Result

A complete LM555 astable multivibrator PCB project was designed, simulated, routed, verified and prepared for manufacturing.